QR Code Registration: dao_client + GAS Implementation Plan

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Status: Draft — awaiting governor approval

1. Problem Statement

There is no way to register a single Agroverse QR code from the autopilot or dao_client. The existing flow requires:

- Adding a row to the "Agroverse QR codes" Google Sheet manually
- Running the batch compiler from a macOS machine with the right fonts and templates

This means the autopilot cannot mint a QR code for an event (like SF Tech Fest) without manual sheet editing. The gap is: **no GAS endpoint for single QR registration** and **no dao_client command to call it**.

2. Solution Architecture

```

dao_client register_qr_code

■

▼

Edgar (sentiment_importer)

■ POST /dao/qr_code_register

■ (signed contribution → Telegram Chat Logs)

▼

GAS Web App (process_qr_code_generation_telegram_logs.gs)

■ Reads Telegram Chat Logs

■ Creates row in "Agroverse QR codes" sheet

■ Triggers GitHub Actions webhook

▼

GitHub Actions (lineage-assets)

■ Runs batch_compiler.py

■ Generates branded QR PNG

■ Commits PNG + JSON manifest

▼

QR code live at truesight.me/qr/?id=<code>
```

```

### Component Breakdown

| Component | Repo | What it does |
|---|---|---|
| **dao_client command** | `dao_client` | New `register_qr_code` CLI command. Signs a `[QR CODE REGISTRATION]` event, submits to Edgar. |
| **Edgar endpoint** | `sentiment_importer` | New `DaoController#register_qr_code` action. Receives signed event, appends to Telegram Chat Logs. |
| **GAS processor** | `tokenomics` (GAS script `1N6o00N9VtRK…`) | New `processQrCodeRegistrationTelegramLogs()` function. Reads `[QR CODE REGISTRATION]` messages, creates single row in Agroverse QR codes sheet. |
| **GitHub Actions** | `lineage-assets` | Existing `generate_qr_batch.sh` workflow. Triggered by GAS webhook. Generates branded QR PNG + manifest. |

## 3. Implementation Phases

### Phase 1: GAS Endpoint (Single QR Registration)

**PR1 — `tokenomics` — Add `register_single_qr_code.gs`**

New GAS file that adds a `doGet(e)` handler for `?action=registerSingleQRCode` with parameters:

- `qr_code` (string) — the QR code ID (e.g. `SFTF_FR_20260612_1`)
- `landing_page` (string) — the URL the QR points to
- `farm_name` (string) — display name
- `state` (string)
- `country` (string)
- `year` (string) — harvest year
- `currency` (string) — product name
- `status` (string) — defaults to `SAMPLE`
- `manager` (string) — who minted it
- `creation_date` (string) — YYYYMMDD

**Logic:**
```

1. Validate required fields
2. Check for duplicate `qr_code` in sheet
3. Append row to "Agroverse QR codes" tab
4. Trigger GitHub Actions webhook for single QR generation
5. Return success/error JSON

Deployment: clasp mirror `1N6o00N9VtRK_L3e0NQXEsmC6QME1KObZdmdbJgo0Tbgj_7P-ElNL5THn`

Phase 2: Edgar Endpoint

PR2 — `sentiment_importer` — Add `DaoController#register_qr_code`

New action that:

1. Accepts POST with signed `[QR CODE REGISTRATION]` event
2. Verifies digital signature
3. Appends to "Telegram Chat Logs" sheet
4. Triggers `WebhookTriggerWorker` → GAS `processQrCodeRegistrationTelegramLogs`

Phase 3: dao_client Command

PR3 — `dao_client` — Add `register_qr_code` CLI command

```bash
truesight-dao-register-qr-code \
--qr-code "SFTF_FR_20260612_1" \
--landing-page "https://agroverse.shop/friends-of-the-rainforest" \
--farm-name "SF Tech Fest" \
--state "CA" \
--country "USA" \
--year "2026" \
--currency "Friends of the Rainforest" \
--status "SAMPLE" \
--manager "Gary Teh"
```

Logic:

1. Build `[QR CODE REGISTRATION]` event payload
2. Sign with contributor key
3. POST to Edgar `/dao/qr_code_register`

4. Print result + QR code URL

Phase 4: Autopilot Integration

PR4 — `truesight_autopilot` — Wire autopilot to use new command

Update the autopilot's QR code workflow so that when a governor asks to mint a QR code:

1. Call `dao_client register_qr_code` with the parameters
2. Wait for the GAS + GitHub Actions pipeline
3. Report the QR code URL back

4. Gates

Gate	Phase	Description	Who
G1	Phase 1	GAS deploys, manual test with curl	Gary
G2	Phase 2	Edgar endpoint deployed, test with signed request	Gary
G3	Phase 3	dao_client command works end-to-end	Gary
G4	Phase 4	Autopilot successfully mints a QR for SF Tech Fest	Gary

5. UAT (User Acceptance Tests)

U1: GAS Single QR Registration

1. `curl` the GAS web app with `?action=registerSingleQRCode&qr_code=UAT_TEST_001&...`
2. Verify row appears in "Agroverse QR codes" sheet
3. Verify status is `MINTED`

U2: Duplicate Detection

1. Register the same QR code twice
2. Verify second call returns error "QR code already exists"

U3: Edgar Endpoint

1. POST signed `[QR CODE REGISTRATION]` to Edgar
2. Verify row appears in "Telegram Chat Logs"

3. Verify GAS processes it and creates sheet row

U4: dao_client Command

1. Run `truesight-dao-register-qr-code --dry-run`
2. Verify output shows correct payload
3. Run without `--dry-run`
4. Verify QR code appears in sheet

U5: End-to-End: SF Tech Fest QR

1. Run dao_client to register `SFTF_FR_20260612_1`
2. Wait for GitHub Actions to generate branded PNG
3. Verify PNG at `lineage-assets/pngs/SFTF_FR_20260612_1.png`
4. Verify JSON manifest at `lineage-assets/qrs/SFTF_FR_20260612_1.json`
5. Verify `https://truesight.me/qr/?id=SFTF_FR_20260612_1` resolves

U6: Error Handling — Missing Fields

1. Call endpoint without required fields
2. Verify descriptive error message

U7: Error Handling — Invalid Status

1. Call with invalid status value
2. Verify validation error

U8: Batch Compatibility

1. Verify existing batch QR generation still works
2. Verify no regression in `processQRCodeGenerationTelegramLogs`

U9: Autopilot Integration

1. Ask autopilot to "mint a QR code for SF Tech Fest"
2. Verify autopilot calls dao_client
3. Verify QR code is generated

6. Files to Create/Modify

New Files

File	Repo	Purpose
`google_app_scripts/agroverse_qr_codes/register_single_qr_code.gs`	`tokenomics`	GAS endpoint for single QR registration
`dao_client/commands/register_qr_code.py`	`dao_client`	CLI command
`sentiment_importer/app/controllers/qr_code_registration_controller.rb`	`sentiment_importer`	Edgar endpoint

Modified Files

File	Repo	Change
`dao_client/setup.py`	`dao_client`	Register new command entry point
`sentiment_importer/config/routes.rb`	`sentiment_importer`	Add route for QR registration
`truesight_autopilot/app/qr_workflow.py`	`truesight_autopilot`	Wire autopilot to use new command

7. Risks & Mitigations

Risk	Likelihood	Mitigation
GAS web app quota exceeded	Low	Single QR registration is lightweight
GitHub Actions webhook fails	Medium	GAS retries with exponential backoff
Duplicate QR code IDs	Low	Sheet-level unique check in GAS
Font/Helvetica missing on autopilot	Medium	Install fonts or use fallback in batch_compiler

8. Timeline (Estimated)

Phase	Effort	Dependencies
Phase 1: GAS	2-3 hours	clasp access to `1N6o00N9VtRK…`
Phase 2: Edgar	1-2 hours	sentiment_importer deploy
Phase 3: dao_client	1-2 hours	dao_client release
Phase 4: Autopilot	1 hour	truesight_autopilot deploy
Total	**5-8 hours**	

9. RESUME HERE

When governor says "go for it":

1. Open PR1: Create `register_single_qr_code.gs` in `tokenomics/google_app_scripts/agroverse_qr_codes/`
2. Deploy GAS via clasp mirror
3. Open PR2: Add Edgar endpoint in `sentiment_importer`
4. Open PR3: Add dao_client command
5. Open PR4: Wire autopilot
6. Run UAT U1–U9
7. Report results in this thread

Gate: PRs are opened only — NEVER self-merge. Governor reviews and merges.